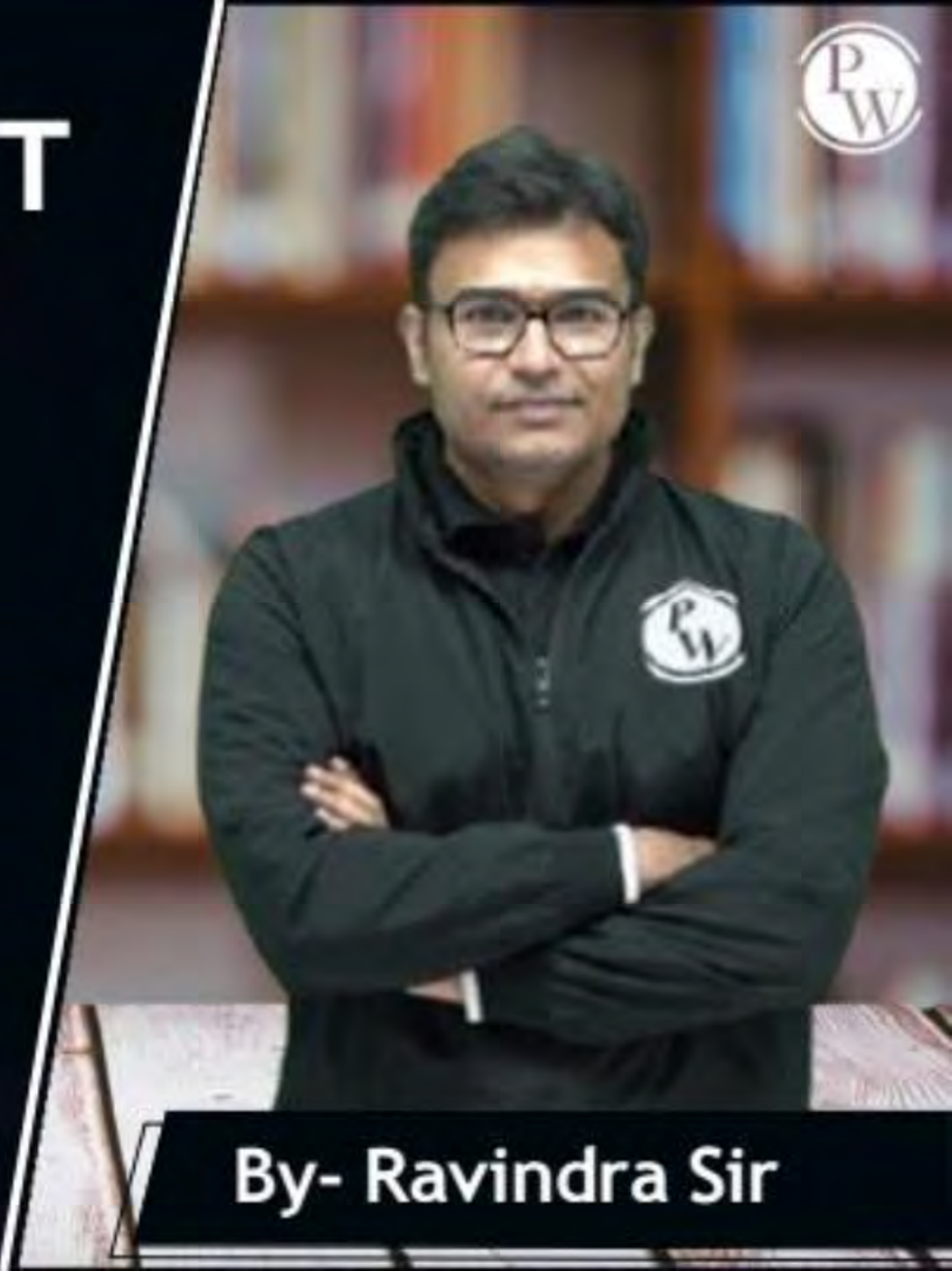
Computer Science & IT

COMPUTER NETWORK
(CN)

ISO-OSI Stack

Lecture No. 07



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

Fragmentation

Topics to be Covered



- Topic
- Topic
- Topic

Reassembly algorithm

Inspiring Stories : Tulir Thapa



Background: Grew up in a small village in Nepal.

Education: Limited schooling.

Achievements: Built handmade wind turbines from junk to light homes.

Impact: Gave electricity to his community without government help.

Inspiring Stories : Sindhi Devi



Background: Rural women from Kutch, Gujarat.

Education: No formal schooling.

Achievements: Revived traditional embroidery into a business.

Impact: Gave income to hundreds of women artisans.

Inspiring Stories : Tulsi Virani



Background: Daughter of a farmer in drought-hit Gujarat.

Education: Finished studies against odds.

Achievements: Created a low-cost water-saving irrigation device.

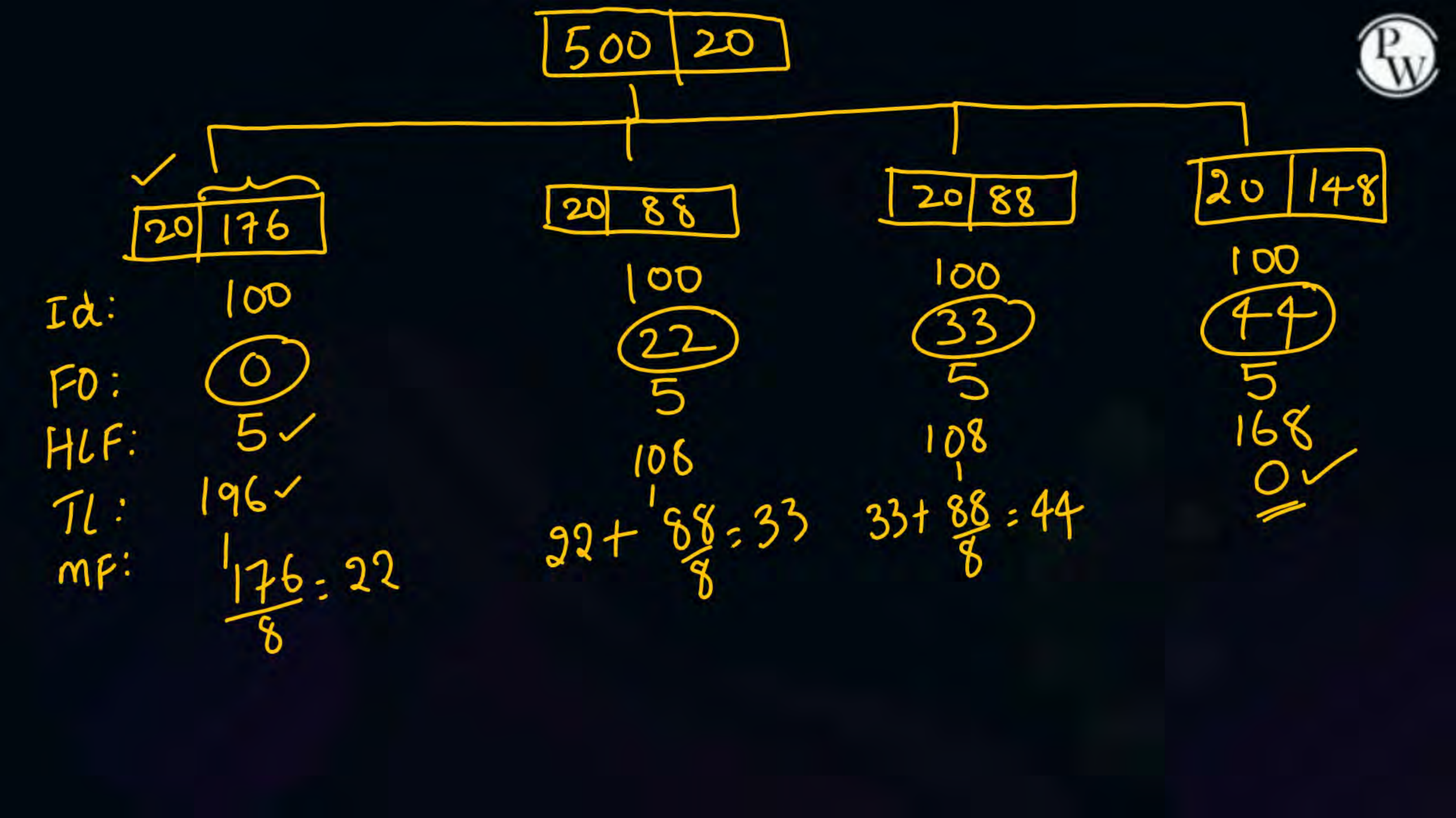
Impact: Helped farmers reduce water wastage and increase crop yield.

Reassembly algorithm

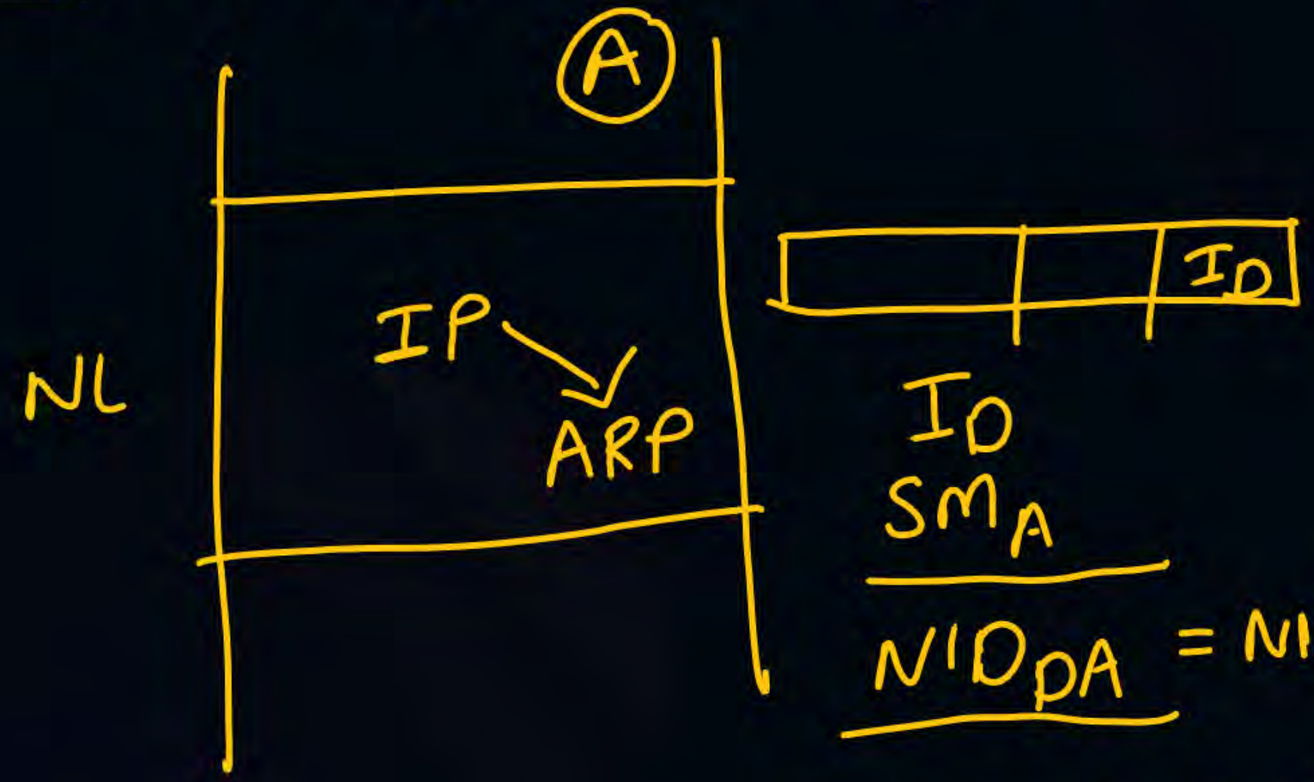
→ Destination should identify that fragmentation is done.

	MF	offset	
fragmentation is done	1	0	— 1 fragment
	1	!0	— Intermediate Fragment
	0	!0	— Last fragment
fragmentation is not done	0	0	— No fragmentation

1) Destination should identify that fragmentation is done (MF, offset)
2) Destination should identify all fragment of a data (Identification number)
3) Identify first fragment (offset = 0)
4) Identify the subsequent fragment → example
5) Repeat step 4, until MF = 0.


500 | 20
20 | 176
20 | 88
20 | 88
20 | 148
Id: 100 100 100 100
FO: 0 22 33 44
HLF: 5 5 5 5
TL: 196 168
MF: 176/8 = 22
22 + 88/8 = 33
33 + 88/8 = 44

ARP: Address resolution protocol



I_D
SM_A

$NID_{DA} = NID_A$

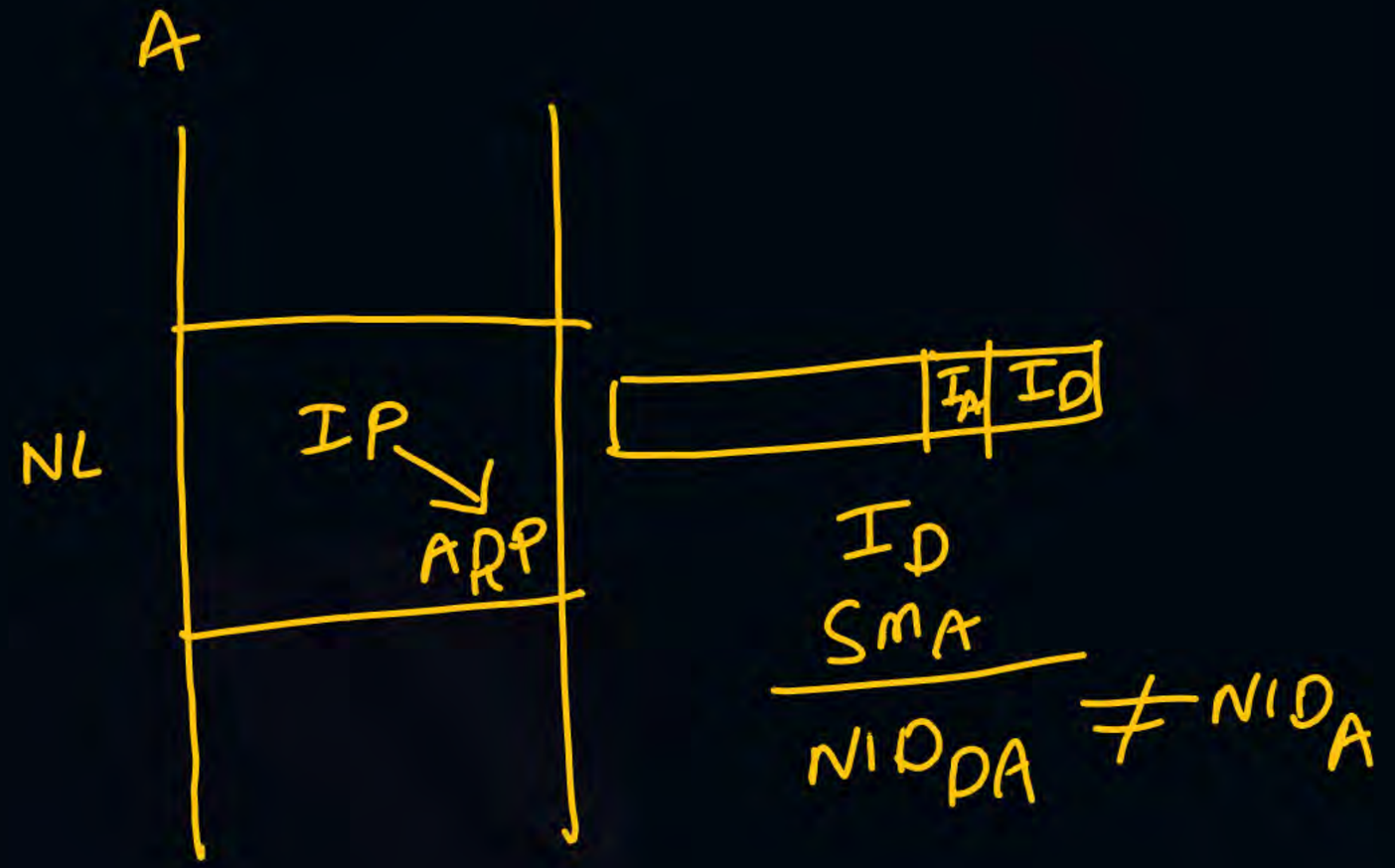


ARP request



ARP Req → Broadcast

ARP Reply → Unicast


A
NL
IP
ARP
I_D
S_{MA}
$NID_{DA} \neq NID_A$


A
I
D
DMAC
IR1
MR1
ARP Req - BC
mac
IR1
MA
ALL
SMAC
DMAC ARP Reply - UC
IR1
MR1
MR1
MA

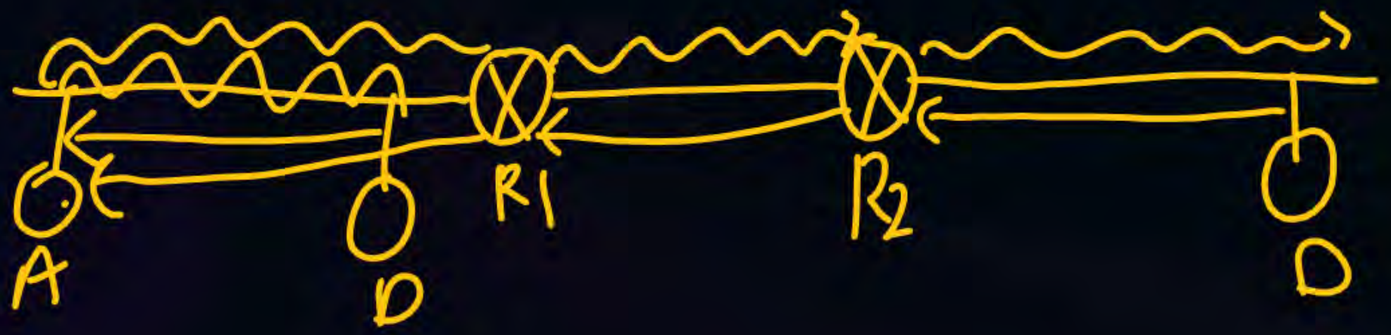
ARP is used in following Situations

(i) Host wants to find MAC addrss of another host. (Same N/w)

(ii) Host wants to find MAC of Router

(iii) Router wants to find the MAC addrss of other router

(iv) Router wants to find the MAC addrss of a host.



THANK - YOU